GAZELIFT: GAZE-GUIDED FEW-SHOT ADAPTATION FOR MEDICAL IMAGE SEGMENTATION

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ABSTRACT

Expert gaze guides target localization but does not delineate boundaries, particularly in low-contrast medical images. Training-free feature refinement does not learn task-specific transformations of pretrained representations. We propose **GazeLIFT**, a gaze-guided, parameter-efficient adaptation framework that combines semantic conditioning with image-detail reconstruction. Expert fixations guide foreground–background feature contrast across multiple depths of a frozen DINOv3 encoder, while a learnable lifting-wavelet transform followed by fixed average pooling provides complementary spatial detail during decoding. Experiments on Kvasir-SEG and NCI-ISBI show that GazeLIFT outperforms the evaluated baselines, achieving 93.35% and 89.24% mean Dice, respectively, with only 0.818 million trainable parameters, excluding the frozen encoder. With nine Kvasir-SEG or six NCI-ISBI training masks, GazeLIFT achieves 83.16% and 80.85% Dice, respectively, supporting its use under limited training-mask supervision.

***Index Terms*—** Medical image segmentation, expert gaze, vision foundation models, few-shot adaptation.

1. INTRODUCTION

Accurate medical image segmentation requires pixel-wise annotations that demand substantial expert effort. Supervised approaches such as U-Net and nnU-Net achieve strong performance [1, 2], but their adaptation to new tasks depends on suitable annotated data. Self-supervised vision models offer an alternative by learning transferable visual representations from large-scale unlabeled images, reducing the need to learn features from scratch. Recent models such as DINOv3 provide rich representations [3], whose potential for medical image segmentation has been previously demonstrated by Dino U-Net [4].

In parallel, previous studies have clearly showed the potential of expert gaze as spatial guidance for medical image segmentation. GazeMedSeg [5] and GradTrack [6] use gaze for weak supervision, whereas GazeSAM [7] and GazeMedSAMv2 [8] use it for spatial prompting. GazeRefine [9] constructs gaze-anchored prototypes in frozen DINOv3 features for training-free segmentation. Although these methods establish the value of gaze, sparse fixations do not delineate boundaries.

To address these limitations, we introduce **GazeLIFT**, a parameter-efficient framework combining gaze-conditioned semantic adaptation with image-detail reconstruction. Our contributions are: **(i) Gaze-conditioned semantic adaptation:** learned multi-depth fusion of projected DINOv3 features, fixation heatmaps and foreground–background contrast, with the encoder frozen. **(ii) Complementary image detail:** an eight-parameter lifting-wavelet transform and fixed pooling supply decoder detail; removing this branch lowers NCI-ISBI Dice by 1.61 percentage points. **(iii) Few-shot target-task adaptation:** experiments use nine Kvasir-SEG images or six NCI-ISBI slices in the 1% training-mask setting, and examine training fractions under matched update budgets, component ablations and backbone size.

2. PROPOSED METHOD

The proposed GazeLIFT addresses two complementary challenges of gaze-guided segmentation: adapting pretrained representations to the target task and recovering spatial details that sparse fixations cannot explicitly describe. As illustrated in Fig. 1, the framework combines gaze-conditioned semantic adaptation with a learnable wavelet pathway, integrating both representations through a lightweight segmentation decoder.

Gaze-conditioned semantic adaptation. Rather than relying solely on training-free feature refinement, GazeLIFT learns task-specific transformations of pretrained representations while keeping the encoder frozen. Given an image $I \in \mathbb{R}^{3\times512\times512}$, frozen pretrained DINOv3 ViT-L/16 provides features $F_s \in \mathbb{R}^{1024\times32\times32}$ from blocks 6, 12, 18 and 24. Sampling these depths exposes the decoder to representations from different encoder stages. Separate learned 1×1 convolutions adapt and compress the features to $c_s \in \{32, 64, 128, 256\}$ channels, followed by eight-group normalization and GELU. Group normalization avoids dependence on batch statistics. Fixed bilinear resizing produces P_s on grids of side length $r_s \in \{256, 128, 64, 32\}$, respectively, aligning the semantic features with the decoder resolutions. Coarser grids receive more channels; the wavelet pathway supplies local image detail.

To introduce expert guidance, we convert sparse fixations into a duration-weighted Gaussian heatmap, corresponding to the gaze-generation branch in Fig. 1. Let $(u, v) \in \Omega = \{0, \dots, 511\}^2$ denote horizontal and vertical pixel coordinates. For J normalized fixations (x_j, y_j), their input-grid positions are $q_j = (q_j^x, q_j^y) = (511x_j, 511y_j)$.

The weight w_j is obtained by min–max normalizing fixation durations within each image. When all durations are equal, we assign $w_j = 1$ to every fixation. We define

$$A(u,v) = \sum_{j=1}^{J} w_j \exp\left[-\frac{(u-q_j^x)^2 + (v-q_j^y)^2}{2\sigma^2}\right], \qquad H(u,v) = \frac{A(u,v)}{\max_{(a,b)\in\Omega} A(a,b)}. \tag{1}$$

Here (a, b) ranges over all image pixels to determine the normalization maximum. The Gaussian bandwidth σ controls the spatial spread around each fixation and is measured in resized input pixels. The reported baseline uses $\sigma = 32$ for Kvasir-SEG and 8 for NCI-ISBI. Average pooling produces H_s at each semantic resolution, aligning gaze with P_s.

Since gaze identifies attended regions rather than complete object boundaries, directly using the heatmap provides limited infor-

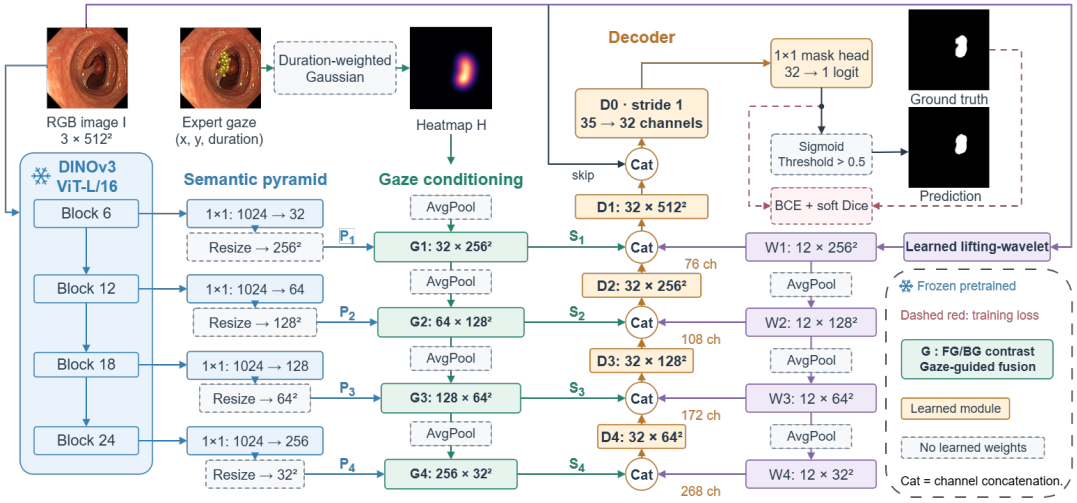


Fig. 1. GazeLIFT architecture: gaze-conditioned DINOv3 semantics and pooled wavelet detail are fused during decoding. Snowflakes mark frozen weights; dashed blocks have no learned weights. Ground truth supervises training only.

mation about the surrounding visual context. We summarize gaze-associated features and their spatial complement using soft foreground and background candidate prototypes. At level s, $\Omega_s = \{0, \dots, r_s - 1\}^2$ denotes the feature grid, and $P_s(u, v) \in \mathbb{R}^{c_s}$ is the feature vector at location (u, v). The prototypes are

$$\mu_s^{\mathrm{FG}} = \frac{\sum_{(u,v)\in\Omega_s} H_s(u,v) P_s(u,v)}{\sum_{(u,v)\in\Omega_s} H_s(u,v)},$$
$$\mu_s^{\mathrm{BG}} = \frac{\sum_{(u,v)\in\Omega_s} [1 - H_s(u,v)] P_s(u,v)}{\sum_{(u,v)\in\Omega_s} [1 - H_s(u,v)]}. \tag{2}$$

These prototypes summarize features associated with expert attention and its spatial complement, without assuming that gaze provides ground-truth segmentation labels.

To extend gaze guidance beyond the fixation locations, we compare each feature vector with both prototypes:

$$C_s(u,v) = \mathrm{ReLU}\left[\cos(P_s(u,v), \mu_s^{\mathrm{FG}}) - \cos(P_s(u,v), \mu_s^{\mathrm{BG}})\right]. \tag{3}$$

For numerical stability, the implementation floors normalization denominators and disables contrast when either candidate has negligible weight. The resulting map identifies locations whose representations are more similar to gaze-associated features than to their complement, allowing semantic guidance to extend beyond the directly observed fixation regions.

Finally, the original projected features, spatial gaze heatmap, and contrast map are combined through a trainable conditioning layer:

$$S_s = \mathrm{GELU}\left(\mathrm{GN}\left(\mathrm{Conv}_{1\times1}[P_s, H_s, C_s]\right)\right). \tag{4}$$

This operation learns how to combine visual information, expert localization, and relative feature similarity for the segmentation task. Gradients propagate through the conditioning and prototype computations to the learned projections, while DINOv3 remains frozen.

Wavelet detail and segmentation decoding. Although gaze-conditioned semantic representations provide target-related information, sparse fixations do not explicitly describe object contours. Moreover, pretrained patch-level features may not retain all the spatial information required for precise segmentation. To complement these representations, GazeLIFT introduces a direct image pathway based on a learnable lifting-wavelet transform [10].

A single separable lifting transform is applied to RGB using shared prediction and update filters initialized to Haar. Eight trainable scalars parameterize the transform. Four subbands per color channel, scaled by $1/2$, produce $W_1 \in \mathbb{R}^{12\times256\times256}$. Three fixed 2×2 average-pooling operations generate W_2, W_3 and W_4 at resolutions 128^2, 64^2 and 32^2, respectively. Thus, only the initial lifting transform is learned; the subsequent detail pyramid uses parameter-free pooling.

At the deepest grid, concatenating S_4 and W_4 produces 268 channels. Successive decoder inputs contain 268, 172, 108 and 76 channels, combining the previous decoder output, the corresponding conditioned semantics and the 12-channel detail features where available. Each unit applies a stride-2 3×3 transpose convolution to 32 channels, GN and GELU, followed by a 3×3 convolution with GN and GELU. At full resolution, concatenating the decoder output with RGB produces 35 channels. The final unit uses stride 1 and retains 32 output channels; a 1×1 head then predicts one segmentation logit per pixel.

The adaptation network contains 817,993 trainable parameters, excluding the frozen DINOv3 encoder, which remains part of inference.

3. EXPERIMENTAL SETUP

Datasets and training. We evaluate GazeLIFT on Kvasir-SEG polyp images [11] and NCI-ISBI prostate MRI [12], using expert fixations from GazeMedSeg [5]. The training/validation/test splits contain 810/90/100 images and 599/73/117 slices, respectively, with patient-disjoint MRI partitions (45/6/8 patients). Images are resized to 512×512 and ImageNet-normalized; MRI slices are min–max normalized and replicated to RGB. We train the adaptation modules using AdamW (2×10^{-4}, weight decay 10^{-4}), effective batch size eight, and BCE plus soft Dice loss. Training runs for up to 80 epochs with cosine learning-rate decay and validation-based checkpoint selection. The frozen DINOv3 encoder is shared across experiments, and expert gaze is required at inference.

Evaluation protocol. We report mean Dice (%) and evaluate component and backbone ablations under shared data splits and training settings. A matched nnU-Net + gaze baseline receives the fixation heatmap as an additional input channel and is trained from scratch under the same supervision and optimization protocol. For reduced-mask experiments, nested training subsets range from 1% to

100%, with fixed validation cohorts (90/73 masks) and test cohorts (100/117 masks). At 1%, only nine Kvasir-SEG or six NCI-ISBI training masks are used. Each fraction receives a matched optimizer-update budget, with validation-based checkpoint selection. Validation masks remain additional supervision, and expert gaze is required at inference.

4. RESULTS

4.1. Quantitative Results

Table 1 compares GazeLIFT with ten baseline configurations spanning full, weak and gaze-guided supervision. GazeLIFT achieves mean Dice scores of 93.35% on Kvasir-SEG and 89.24% on NCI-ISBI. Under the matched training protocol, it improves upon nnU-Net + gaze by 2.76 and 1.09 percentage points, respectively, using 0.818M trainable parameters versus 30.01M. Literature results provide additional context, although their supervision and evaluation protocols differ.

Table 1. Quantitative comparison on Kvasir-SEG and NCI-ISBI: mean Dice (%) ± SD. Literature results retain their reported protocols and SDs; fully and weakly supervised results follow GradTrack [6]. Matched nnU-Net + gaze and GazeLIFT results. Bold indicates the higher matched mean. ZS: zero-shot.

Method	Supervision	Kvasir	NCI-ISBI
nnU-Net [2]	Full	88.41±0.47	82.43±0.18
AGMM [13]	Point	75.57±0.84	73.86±1.26
DMPLS [14]	Scribble	69.23±0.32	57.44±0.56
BoxTeacher [15]	Box	73.33±1.30	75.60±1.15
GazeMedSeg [5]	Gaze	77.80±1.02	77.64±0.57
GradTrack [6]	Gaze	81.01±0.66	80.25±0.40
GazeSAM [7]	ZS + gaze	80.43±0.22	82.60±0.17
GazeMedSAMv2 [8]	ZS + gaze	85.19±0.13	85.62±0.10
GazeRefine [9]	ZS + gaze	88.10±0.14	80.28±0.11
nnU-Net + gaze [2]	Full + gaze	90.59±0.12	88.15±0.11
GazeLIFT (ours)	Full + gaze	**93.35±0.08**	**89.24±0.10**

4.2. Ablation Study

Main component contributions. Table 2 separates the contribution of gaze from the conditional benefits of the proposed modules. *Without expert gaze* lowers Dice by 3.30/6.14 pp on Kvasir-SEG/NCI-ISBI, establishing gaze's effect within the full architecture. *Without wavelet information* retains gaze guidance but lowers NCI-ISBI Dice by 1.61 pp, supporting a complementary contribution from image detail; the Kvasir-SEG change is small. *Last DINO depth only* lowers NCI-ISBI Dice by 1.94 pp but slightly improves Kvasir-SEG, indicating a dataset-dependent benefit of multi-depth semantics. Duration weighting and BG subtraction produce small changes. The *DINO-only + decoder* comparison measures the combined addition of gaze, wavelet detail and RGB, rather than gaze alone.

Statistical analysis across training runs Across 15 paired training runs (seeds 40–54), GazeLIFT achieves achieves $93.22 \pm 0.44\%$, compared with $90.28 \pm 0.54\%$ without gaze. On NCI-ISBI, the corresponding results are $87.94 \pm 0.79\%$ and $84.25 \pm 0.88\%$. The mean paired improvements are 2.94% on Kvasir-SEG (95% CI: [2.69, 3.18]) and 3.70% on NCI-ISBI (95% CI: [3.11, 4.29]). Both paired confidence intervals exclude zero, supporting a positive average contribution of gaze on the evaluated cohorts. GazeLIFT also exhibits lower observed between-run SD on both datasets, providing evidence of repeatability across training seeds.

Table 2. Ablation results reported as mean Dice (%) ± SD of per-case Dice coefficients on Kvasir-SEG and NCI-ISBI.

Configuration	Kvasir	NCI-ISBI
Full model	93.35±0.08	89.24±0.10
Without expert gaze	90.05±0.15	83.11±0.17
Unweighted gaze	93.49±0.09	89.21±0.10
FG maps; no BG subtraction	93.25±0.09	89.18±0.09
Without wavelet information	93.31±0.08	87.63±0.11
Without final RGB skip	92.99±0.10	88.06±0.12
Without DINO and its maps	88.66±0.10	88.67±0.08
DINO-only + decoder	89.78±0.17	83.86±0.16
Last DINO depth only	93.68±0.08	87.31±0.12

Backbone and adaptation efficiency. Table 3 compares active parameter counts. GazeLIFT uses 0.818M parameters, versus 5.11M for Dino U-Net S and 30.01M for nnU-Net + gaze; DINO-based counts exclude frozen encoders. The Small, Base and Large DINOv3 backbones achieve 91.67%, 92.79% and 93.35% Dice on Kvasir-SEG, and 86.86%, 88.59% and 89.24% on NCI-ISBI, respectively.

Figure 3 evaluates adaptation under reduced training-mask supervision with matched optimizer-update budgets. Using only nine Kvasir-SEG or six NCI-ISBI training masks (1%), GazeLIFT achieves 83.16% and 80.85%, respectively. At 50% supervision, performance increases to 91.53% and 87.43%, approaching the full-data results of 93.35% and 89.23%. Validation sets remain fixed, and expert gaze is required at inference.

Table 3. Active parameters (millions). Literature counts follow Dino U-Net [4]; nnU-Net + gaze is measured from its evaluated configuration. DINO-based counts exclude frozen encoders.

Method	Active M	Method	Active M
nnU-Net [2]	30.45	SegResNet [16]	10.26
UNet++ [17]	14.39	U-Mamba [18]	63.76
U-KAN [19]	9.38	Swin-UMamba† [20]	37.76
SAM2-UNet [21]	5.38	Dino U-Net S [4]	5.11
Dino U-Net B [4]	11.65	Dino U-Net L [4]	18.14
Dino U-Net 7B [4]	228.97	nnU-Net + gaze [2]	30.00
GazeLIFT (ours)	**0.82**		

4.3. Qualitative Results

Figure 2 illustrates segmentation results on selected Kvasir-SEG and NCI-ISBI samples and are compared to state-of-the-art methods. GazeLIFT shows closer agreement with the annotated polyp contours, whereas competing methods exhibit over-segmentation. On prostate MRI, GazeLIFT provides more complete target coverage, particularly along the lower boundary, where competing predictions omit portions of the prostate. These examples complement the quantitative results by illustrating differences in target localization and boundary alignment.

5. CONCLUSION

GazeLIFT combines gaze-conditioned frozen DINOv3 features with learnable wavelet detail and compact decoding. With 0.818M trainable parameters, it achieves 93.35% and 89.24% Dice on Kvasir-SEG and NCI-ISBI, respectively. Component ablations highlight the contribution of expert gaze and the dataset-dependent benefits of wavelet detail and multi-depth representations. Experiments using nine and six training masks demonstrate few-shot target-task adaptation, with separate validation supervision and expert gaze at infer-

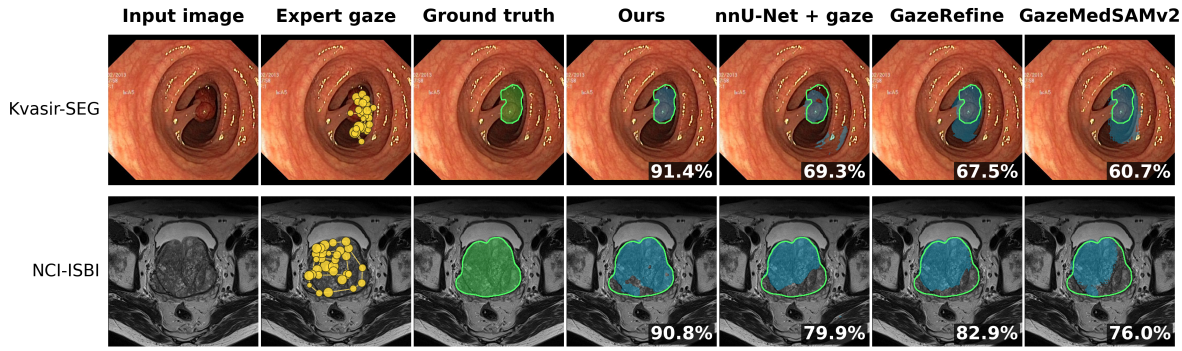


Fig. 2. Qualitative results on one test case per dataset with Dice scores (%) displayed for each prediction. Yellow markers and lines indicate expert gaze, with marker size encoding fixation duration. Green outlines indicate ground truth, and cyan overlays show predicted masks.

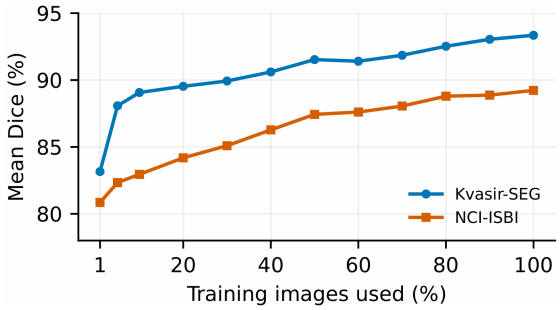


Fig. 3. Full-model adaptation versus training-mask fraction, with fixed updates. At 1%: 9 Kvasir-SEG and 6 NCI-ISBI training masks.

ence. Future work will investigate additional datasets, multi-class and multi-instance segmentation, extension to 3D volumetric images, and gaze-guided training without requiring gaze at inference.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study used existing medical images and expert gaze annotations. No new participants were recruited.

7. ACKNOWLEDGMENTS

8. REFERENCES

[1] Olaf Ronneberger et al., "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention – MICCAI*, 2015, pp. 234–241.

[2] Fabian Isensee et al., "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, pp. 203–211, 2021.

[3] Oriane Siméoni et al., "DINOv3," 2025, arXiv:2508.10104.

[4] Haoyue Li et al., "Dino u-net: Exploiting high-fidelity dense features from foundation models for medical image segmentation," 2026, arXiv:2508.20909v2.

[5] Yuan Zhong et al., "Weakly-supervised medical image segmentation with gaze annotations," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. 2024, vol. 15003, pp. 530–540, Springer Nature Switzerland.

[6] Zhisong Wang et al., "Enjoying information dividend: Gaze track-based medical weakly supervised segmentation," in *MICCAI*, 2025, Open-access accepted paper; arXiv:2505.22230.

[7] Bin Wang et al., "GazeSAM: What you see is what you segment," 2023, arXiv:2304.13844.

[8] Tatyana Shmykova et al., "Zero-shot gaze-based volumetric medical image segmentation," 2025, arXiv:2505.15256.

[9] Mohammed Oussama Benyahia et al., "GazeRefine: Expert gaze as a test-time prompt for training-free medical image segmentation," 2026, arXiv:2609.01310; accepted at a MICCAI 2026 workshop.

[10] Wim Sweldens, "The lifting scheme: A custom-design construction of biorthogonal wavelets," *Applied and Computational Harmonic Analysis*, vol. 3, no. 2, pp. 186–200, 1996.

[11] Debesh Jha et al., "Kvasir-SEG: A segmented polyp dataset," in *MultiMedia Modeling*, 2020, vol. 11962, pp. 451–462.

[12] B. Nicolas Bloch et al., "NCI-ISBI 2013 challenge: Automated segmentation of prostate structures," 2015.

[13] Linshan Wu et al., "Sparsely annotated semantic segmentation with adaptive gaussian mixtures," in *Proc. IEEE/CVF CVPR*, 2023, pp. 15454–15464.

[14] Xiangde Luo et al., "Scribble-supervised medical image segmentation via dual-branch network and dynamically mixed pseudo labels supervision," 2022, arXiv:2203.02106.

[15] Tianheng Cheng et al., "BoxTeacher: Exploring high-quality pseudo labels for weakly supervised instance segmentation," 2023, arXiv:2210.05174.

[16] Andriy Myronenko, "3D MRI brain tumor segmentation using autoencoder regularization," 2018, arXiv:1810.11654.

[17] Zongwei Zhou et al., "UNet++: A nested U-Net architecture for medical image segmentation," 2018, arXiv:1807.10165.

[18] Jun Ma et al., "U-Mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, arXiv:2401.04722.

[19] Chenxin Li et al., "U-KAN makes strong backbone for medical image segmentation and generation," 2024, arXiv:2406.02918.

[20] Jiarun Liu et al., "Swin-UMamba: Mamba-based UNet with ImageNet-based pretraining," 2024, arXiv:2402.03302.

[21] Xinyu Xiong et al., "SAM2-UNet: Segment anything 2 makes strong encoder for natural and medical image segmentation," *Visual Intelligence*, vol. 4, no. 1, pp. 2, 2026.